A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 06

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 6 points below. **1 Location:** discharge into the sea / large water bodies where currents dilute and disperse the pollutant **1 Location:** chimneys / emissions sited downwind of urban areas so pollutants are not carried over populations – avoiding valleys and enclosed water bodies where pollutants concentrate **1 Location:** not discharging wastes onto permeable rock above an aquifer / not siting landfill on aquifers **1 Timing:** restrict polluting activities (industrial combustion, vehicle use) during temperature inversions when dispersal is poor **1 Timing:** discharge liquid effluent when river flow is high / on an ebb tide so it is diluted and carried away **1 Timing:** pesticide spraying at night / on still days – noisy operations restricted to daytime – night-flight restrictions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **permeable** → how easily water passes through / lets water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	6
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** The organisation / person that causes the pollution is responsible for (paying for) the damage caused **1** If the cost of preventing the pollution is less than the cost of the damage / fines, there is a financial incentive to prevent it – works best when penalties are high and escape is unlikely **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** It is assumed that the waste will cause pollution / harm until research shows it is unlikely to **1** So release is not permitted until it has been tested – lack of knowledge of a problem is not an excuse / does not reduce responsibility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.	

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Question	Marking guidance	Marks
02.1	1 mark for the point below. **1** The time taken for half of the atoms of a radioactive isotope to decay (activity halves) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	1
02.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 32 / 8 = 4 half-lives **1** 6400 / 2^4 = 400 Bq **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Bq) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient **1** Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low **1** Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3

Do not write outside the box

Question	Marking guidance	Marks
02.4	**1 mark each** 1 mark for each of the 3 points below. **1** Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure **1** Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) **1** Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	3
02.5	1 mark for the point below. **1** A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ionising radiation** → radiation with enough energy to remove electrons and form ions	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** B, C and D – each loses soil faster than the 1 t ha^{-1} yr^{-1} at which it forms, so the soil is getting shallower **1** Accelerated erosion means the natural equilibrium between soil formation and erosion has been broken – field A loses only 0.8 t ha^{-1} yr^{-1}, which is less than the rate of formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (24.0 - 9.0) / 24.0 x 100 **1** = 62.5% (accept 62 - 63%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. • **contour ploughing** → ploughing along the slope rather than up and down it **Do not credit** • A bare wrong answer with no working.	2
03.3	**1 mark each** 1 mark for each of the 3 points below. **1 Contour ploughing:** furrows run along the contour at 90° to the slope so runoff is stopped by the furrow and loses kinetic energy – the particles it was carrying are deposited in the furrow instead of being washed away **1 Terracing:** the slope is cut into a series of narrow level fields held by retaining walls, so water slows as it crosses each field and much of it infiltrates rather than running off **1 Rows of stones:** stones laid along the contours of a gentle slope slow the runoff and trap disturbed soil behind them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Contour ploughing** → ploughing along the slope rather than up and down it • **Terracing** → cutting the slope into level steps	3

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1 Multicropping:** growing more than one crop in the field at a time, sown and harvested at different times, so some crop always covers the soil **1 Strip cropping:** wide strips of different crops, most effective at 90° to the prevailing wind, for farms that use machinery **1** Growing a long-term crop such as permanent grassland or a bush or tree crop, which does not need frequent replanting or soil disturbance **1** No-tillage cultivation / direct drilling of seed, which disturbs the soil far less than ploughing and sowing **1** Adding organic matter such as mulch or manure – surface mulch shields the soil from the wind and from rain splash, and the humus formed as it decomposes makes the particles adhere to each other **1 Controlling livestock:** lower stocking density and restricted movement so there is less trampling damage and more plant cover **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
03.5	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Soil particles carried into rivers are deposited where the water slows, so the flow capacity of the river falls and it overflows more readily after heavy rain **1** Sediment fills reservoirs, reducing the volume of water they can store and shortening their useful life **1** High turbidity in the river reduces light penetration and smothers the river bed, damaging aquatic life **1** Wind erosion raises the level of atmospheric particulates, which worsens respiratory illness such as asthma **1 Erosion contributes to desertification:** vegetation struggles to grow on the thinner soil, which exposes more soil and reduces rainfall further **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **particulates** → fine solid particles / smoke and dust **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-166; AQA spec 3.2.5.2.	

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Question	Marking guidance	Marks
04.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** Σn(n - 1) = (40 x 39) + (30 x 29) + (20 x 19) + (10 x 9) = 1560 + 870 + 380 + 90 = 2900 **1** N(N - 1) = 100 x 99 = 9900 **1** D = 9900 / 2900 = 3.41 (accept 3.4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Simpson** → a diversity index that uses both the number of species and their abundance **Do not credit** • A bare wrong answer with no working. • 'more species always means a higher index' - the evenness of the abundances matters as well	3
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Woodland A has a higher biodiversity (greater variety and more even abundance) than woodland B **1** The index does not identify which species are present / a rare or important species may be missed – depends on identification skill and sampling method **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402; AQA spec 3.7.2.3.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1** 1 nitrogen fixation – 2 nitrification **1** 3 denitrification – 4 ammonification **1** 5 ionisation – 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **nitrification** → ammonium oxidised to nitrite and then nitrate • **denitrification** → nitrate converted back to nitrogen gas	3

Do not write outside the box

Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Drainage makes the soil aerobic **1** Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate **1** Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **nitrifying** → ammonium oxidised to nitrite and then nitrate • **denitrifying** → nitrate converted back to nitrogen gas	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrogen gas is very unreactive so most organisms cannot use it directly **1** Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate – the cycle recycles a small usable amount **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins **1** When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrification** → ammonium oxidised to nitrite and then nitrate	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	
06.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Sludge from primary sedimentation is digested by anaerobic microbes in a warm tank for about four weeks **1** Kills most pathogens **1** Reduces odour and volume of the sludge **1** Produces biogas (mainly methane with some CO2) used as fuel to heat the digester / generate electricity **1** Digested sludge has a better C:N ratio and can be used as agricultural fertiliser **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

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Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** Heavy metals from urban wastes / road runoff may be present and absorbed by crops - reduced by adding lime to lower metal solubility **1** Remaining pathogens - destroyed by heat treatment of the sludge before use **1 Timing:** spreading in the growing season covers crops – in winter vehicles compact wet soils - apply at suitable times / when soils are dry **1** Odour / public objection - digestion reduces odour **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Foul waste is a small volume needing full treatment so smaller, cheaper treatment works can be built – surface water needs little treatment **1** The treatment works is not affected by weather-related fluctuations in flow (storms) so storm overflows of untreated sewage are avoided **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 270-273; AQA spec 3.4.3.2.9.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1
07.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2

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Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
08.1	**1 mark each** 1 mark for each of the 3 points below. **1** 2020 has higher peak flow in the wet season (32 vs 22 $m^3 s^{-1}$ in March) **1** 2020 has much lower dry-season flow (2 vs 9 $m^3 s^{-1}$ in July / August) **1** Flow is much more variable / seasonal in 2020 – the river almost dries up **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** In 1990 snow / ice stored wet-season precipitation and released it gradually as meltwater during the dry season, evening out the flow **1** With less ice, precipitation now falls as rain / melts immediately and runs off quickly, increasing wet-season flow and flood risk **1** Little stored ice remains to melt in the dry season so flow falls very low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
08.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Water shortages for irrigation / drinking / HEP in the dry season **1** Flooding in the wet season **1** Sudden floods from ice-lake outbursts (glacial lake outburst floods) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

Do not write outside the box

Question	Marking guidance	Marks
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surface meltwater flows down cracks to the base, lubricating the ice so it slides faster – if this exceeds the faster melting at the front, the front advances **1** But if snowfall at the source has not increased, total volume falls – both retreating and extending glaciers can indicate warming **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 94-96; AQA spec 3.2.1.2.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** A turbofan at the front forces bypass air around the inner cowling / core jet **1** The bypass air smooths the flow of the turbulent hot exhaust gases, reducing the turbulence that creates noise **1** The higher the ratio of bypass air to jet exhaust the quieter the engine (Trent 1000 quietest, Olympus noisiest) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.2	**Any 2 of 6** 1 mark each for any 2 of the 6 points below, in any order, maximum 2. **1** Chevron (serrated) nozzles mix bypass air with surrounding air more smoothly **1** Acoustic liners in the engine cowlings absorb noise **1** Hush kits fitted to older low-bypass engines **1** Fairings on undercarriage and wing-flap hinges reduce turbulence – welded rather than riveted panels **1** Lighter composite aircraft need less thrust **1** Blended wing with engines on top so the body shields the ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 4 points below. **1** Constant (3°) descent avoids periods of high thrust in a stepped descent and keeps aircraft higher for longer **1** Night flights (especially take-offs, which are noisier) are restricted / banned when people are sleeping and background noise is low **1** Noisier aircraft pay higher charges / use up more quota points and fines for exceeding limits, so airlines are encouraged to buy quieter aircraft **1** More time to slow down after landing so wheel brakes can be used instead of noisy reverse thrusters **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
09.4	1 mark for the point below. **1** Engines run at higher power / maximum thrust and the steep climb angle needs full power (although the aircraft climbs quickly out of earshot) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 288-292; AQA spec 3.4.3.2.13.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Features:** renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts • **Fossil fuels:** high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO2), politically concentrated (Middle East oil) • **Nuclear:** very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition • **Renewables:** low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs • **Sustainability concepts:** impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion) • **Economic and political factors:** subsidies, guaranteed prices, planning, external costs not included in price, energy security • **Judgement:** a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable – transition must start before depletion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **sustainability** → meeting present needs without stopping future generations meeting theirs • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **base load** → the steady minimum demand that must always be met **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	

Do not write outside the box

Question	Marking guidance	Marks
	Model answer - Level 3 (8 marks)	
	Whether a resource suits a use depends on its features: renewable or not, abundance, energy density, locational constraints, intermittency and predictability, the conversions needed, ease of storage and transport, how developed the technology is and its impacts. Fossil fuels dominate because they combine high energy density, easy storage and transport, high temperatures for industry and mature infrastructure - which is why petrol and diesel still monopolise transport. But they are finite, cause acid rain, smoke and CO_2, and are politically concentrated. Nuclear fuel has the highest energy density and gives reliable base load with few deliveries, yet the technology is complex and expensive and produces long-lived waste against public opposition.	
	Renewables have the opposite profile: low energy density needing large collectors, intermittency (wind and solar unpredictable, tidal predictable), strong locational constraints and no direct storage, so they need conversion, storage and grid links and cannot yet reach fossil-fuel temperatures or replace liquid fuels. Sustainability must count impacts before use - extraction, processing, embodied energy of equipment - as well as pollution, habitat damage and depletion during use, and economics and politics (subsidies, planning, external costs left out of prices, energy security) shape what is actually chosen.	
	No single resource is suitable for everything. Electricity, heat and transport have different requirements, so the most sustainable system is a diverse mix matched to each use with storage, and the transition must begin before depletion forces it.	
	Weaker answer - Level 1 (2 marks)	
	Energy resources have different features. Fossil fuels are non-renewable and will run out but they have a high energy density. Renewable energy like wind and solar will not run out but they are intermittent because the wind does not always blow. Nuclear power produces radioactive waste. Renewable energy is more sustainable because it does not produce carbon dioxide. So we should use more renewable energy.	
	What makes the difference	
	• Level 1 mentions two or three features; Level 3 works through the full set and links each to a specific use (transport, base load, industry).	
	• Level 3 includes sustainability concepts (impacts before and during use, external costs) and political factors that Level 1 omits entirely.	
	• The conclusion in Level 3 is a reasoned position (a diverse mix matched to uses); Level 1's 'use more renewables' does not follow from a discussion.	

Do not write outside the box

Question	Marking guidance	Marks
10.2	1 mark for the point below. **1** Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'nuclear power releases no carbon dioxide' - mining, fuel processing, construction and decommissioning all do	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Composition and structure:** N_2, O_2, Ar, CO_2, ozone – troposphere and stratosphere – pressure and liquid water – gases for biological molecules and industry • **Energy processes:** UV absorbed by O / O_2 / O_3 dynamic equilibrium – visible light absorbed at the surface – infrared absorbed by greenhouse gases (33 °C warming) – heat distribution by winds and ocean currents – transport of water vapour – energy budget and albedo • **Global climate change:** anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species – difficulties of monitoring and prediction – control strategies and agreements • **Ozone depletion:** CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • **Other threats:** aerosols, smoke, land-use change – interconnection of processes	25
	• **Evaluation:** which functions are most at risk – success of ozone controls vs slow progress on greenhouse gases – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Context:** uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • **New sources:** rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • **Management:** artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • **Conservation:** low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering – pollution control to protect sources • **Treatment:** matching quality to use – screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	

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